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JEE MAIN PHYSICS DPP: FREE BODY DIAGRAMS & CONNECTED BODIES

Topic: FBD Construction, Pulleys, String Constraints & Connected Systems | Time: 45 Min | Max Marks:

80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** In a standard Atwood machine, two masses $m_1 = 3$ kg and $m_2 = 2$ kg are connected by a light, inextensible string passing over a frictionless, massless pulley. The acceleration of the system is: (Take $g = 10$ m/s²)
- (a) 1.0 m/s²
(b) 2.0 m/s²
(c) 3.0 m/s²
(d) 4.0 m/s²
- Q2.** In the Atwood machine described in Q1, the tension in the string connecting the two masses is:
- (a) 20 N
(b) 30 N
(c) 24 N
(d) 48 N
- Q3.** In Q1, the total downward reaction force exerted by the string on the pulley axle is:
- (a) 24 N
(b) 50 N
(c) 25 N
(d) 48 N
- Q4.** A block of mass $m_1 = 4$ kg rests on a smooth horizontal table and is connected by a light string passing over a frictionless pulley at the table edge to a hanging block of mass $m_2 = 6$ kg. The acceleration of the system is: (Take $g = 10$ m/s²)
- (a) 6.0 m/s²
(b) 4.0 m/s²
(c) 10.0 m/s²
(d) 2.4 m/s²
- Q5.** In the system of Q4, the tension in the connecting string is:
- (a) 60 N
(b) 24 N
(c) 40 N
(d) 36 N
- Q6.** Three blocks of masses $m_1 = 1$ kg, $m_2 = 2$ kg, and $m_3 = 3$ kg are in contact on a smooth horizontal surface. A horizontal force $F = 18$ N is applied to m_1 towards m_3 . The contact force between m_2 and m_3 is:
- (a) 15 N
(b) 12 N
(c) 9 N
(d) 6 N
- Q7.** A block of mass M is pulled along a frictionless horizontal surface by a uniform heavy rope of mass m using a horizontal force F applied at the free end of the rope. The tension at the midpoint of the rope is:
- (a) F
(b) $\frac{MF}{M+m}$
(c) $\frac{mF}{2(M+m)}$
(d) $\frac{(M+m/2)F}{M+m}$
- Q8.** A monkey of mass $m = 40$ kg climbs up a vertical light rope which can withstand a maximum tension of $T_{\max} = 600$ N. The maximum acceleration with which the monkey can safely climb up is: (Take $g = 10$ m/s²)
- (a) 5.0 m/s²
(b) 10.0 m/s²
(c) 2.5 m/s²
(d) 15.0 m/s²
- Q9.** A block of mass $m = 5$ kg is kept on a smooth inclined plane of inclination $\theta = 30^\circ$. The block is connected to a light string passing over a smooth pulley at the top of the incline supporting a hanging mass M . For the system to remain in static equilibrium, the mass M must be:
- (a) 5.0 kg
(b) 2.5 kg
(c) $5\sqrt{3}$ kg
(d) 10.0 kg
- Q10.** In a movable pulley system, a block of mass M is supported by a movable pulley. A person pulls the free end of the string with force F . The acceleration of the block M upward in terms of F , M , and g is: (Assume massless, frictionless pulley)
- (a) $\frac{F}{M} - g$
(b) $\frac{F}{2M} - g$
(c) $\frac{2F}{M} - g$

(d) $\frac{4F}{M} - g$

Q11. A painter of mass $M = 60$ kg stands in a cradle of mass $m = 20$ kg and pulls himself up with a light rope passing over a fixed smooth overhead pulley. To ascend with an acceleration $a = 2 \text{ m/s}^2$, the force with which he must pull the rope is: (Take $g = 10 \text{ m/s}^2$)

- (a) 960 N
- (b) 800 N
- (c) 240 N
- (d) 480 N

Q12. In Q11, the normal contact force exerted by the painter on the floor of the cradle is:

- (a) 240 N
- (b) 480 N
- (c) 720 N
- (d) 120 N

Q13. Two masses $m_1 = 5$ kg and $m_2 = 10$ kg are connected by a light string on a double inclined plane with inclinations $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$ over a smooth apex pulley. The acceleration of mass m_2 down the plane is: (Take $g = 10 \text{ m/s}^2$)

- (a) $\frac{10\sqrt{3}-5}{3} \text{ m/s}^2$
- (b) $\frac{2\sqrt{3}-1}{3} \times 5 \text{ m/s}^2 \approx 4.11 \text{ m/s}^2$
- (c) 2.5 m/s^2
- (d) 5.0 m/s^2

Q14. A block of mass $m = 2$ kg is suspended by two light strings making angles $\theta_1 = 30^\circ$ and $\theta_2 = 60^\circ$ with the horizontal ceiling. The tensions T_1 and T_2 in the strings are: (Take $g = 10 \text{ m/s}^2$)

- (a) $T_1 = 20 \text{ N}$, $T_2 = 10 \text{ N}$
- (b) $T_1 = 10\sqrt{3} \text{ N}$, $T_2 = 10 \text{ N}$
- (c) $T_1 = 10 \text{ N}$, $T_2 = 10\sqrt{3} \text{ N}$
- (d) $T_1 = 10\sqrt{2} \text{ N}$, $T_2 = 10\sqrt{2} \text{ N}$

Q15. In a wedge-block system, a block of mass m is placed on a smooth wedge of inclination θ . To keep the block stationary relative to the wedge, the horizontal acceleration a given to the wedge must be:

- (a) $g \sin \theta$ towards right
- (b) $g \cos \theta$ towards left
- (c) $g \cot \theta$ towards left
- (d) $g \tan \theta$ towards left

Q16. In the previous question, the normal force exerted by the wedge on the block during this accelerated motion is:

- (a) $\frac{mg}{\cos \theta}$
- (b) $mg \cos \theta$
- (c) $mg \sin \theta$
- (d) $mg(1 + \tan \theta)$

Q17. A block A of mass m is attached to a light spring of stiffness k , which is connected to a block B of mass $2m$. The system is placed on a frictionless table. If block A is pulled with acceleration a_0 , the acceleration of block B at that exact instant is:

- (a) a_0
- (b) 0 (until the spring deforms)
- (c) $\frac{a_0}{2}$
- (d) $2a_0$

Q18. A chain of length L and mass M is held vertically with its lower end touching a table. If it is released from rest, the force exerted by the chain on the table when a length x has fallen is:

- (a) $\frac{Mgx}{L}$
- (b) $\frac{2Mgx}{L}$
- (c) $\frac{3Mgx}{L}$
- (d) $\frac{4Mgx}{L}$

Q19. In a pulley-block system, block A moves downwards with speed v_A . If the string is looped around a movable pulley attached to block B , the constraint relation yields the upward velocity v_B as:

- (a) $v_B = 2v_A$
- (b) $v_B = v_A$
- (c) $v_B = 4v_A$
- (d) $v_B = \frac{v_A}{2}$

Q20. A sphere of mass M is held between a vertical wall and an inclined plane of inclination $\theta = 60^\circ$ with the horizontal. Assuming all surfaces are smooth, the normal force exerted by the vertical wall on the sphere is:

- (a) $\frac{Mg}{\sqrt{3}}$
- (b) $\sqrt{3}Mg$
- (c) $2Mg$
- (d) $\frac{Mg}{2}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

 Net driving force: $F_{\text{net}} = (m_1 - m_2)g$.

 Total mass: $M_{\text{total}} = m_1 + m_2$.

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{10}{5} = 2.0 \text{ m/s}^2$$

Sol 2. Correct Option: (c)

Tension in the string:

$$T = \frac{2m_1m_2g}{m_1 + m_2} = \frac{2 \times 3 \times 2 \times 10}{3 + 2} = \frac{120}{5} = 24 \text{ N}$$

Sol 3. Correct Option: (d)

 The pulley supports two vertical segments of string with tension T on either side. Downward thrust on the axle:

$$R_{\text{pulley}} = 2T = 2 \times 24 = 48 \text{ N}$$

Sol 4. Correct Option: (a)

 For the table-block system, m_2 accelerates the entire system:

$$a = \frac{m_2g}{m_1 + m_2} = \frac{6 \times 10}{4 + 6} = \frac{60}{10} = 6.0 \text{ m/s}^2$$

Sol 5. Correct Option: (b)

 Tension in string equals force accelerating m_1 :

$$T = m_1a = 4 \times 6.0 = 24 \text{ N}$$

Sol 6. Correct Option: (c)

System acceleration:

$$a = \frac{F}{m_1 + m_2 + m_3} = \frac{18}{1 + 2 + 3} = 3 \text{ m/s}^2$$

 Contact force between m_2 and m_3 accelerates m_3 :

$$N_{23} = m_3a = 3 \times 3 = 9 \text{ N}$$

Sol 7. Correct Option: (d)

 Acceleration of system: $a = \frac{F}{M + m}$.

 The midpoint of the rope accelerates the mass behind it, which is $\left(M + \frac{m}{2}\right)$:

$$T_{\text{mid}} = \left(M + \frac{m}{2}\right)a = \frac{(M + m/2)F}{M + m}$$

Sol 8. Correct Option: (a)

Equation of motion for upward climbing:

$$T - mg = ma \implies a = \frac{T - mg}{m}$$

$$a_{\text{max}} = \frac{600 - (40 \times 10)}{40} = \frac{200}{40} = 5.0 \text{ m/s}^2$$

Sol 9. Correct Option: (b)

For equilibrium on a smooth incline:

$$T = mg \sin \theta \quad \text{and} \quad T = Mg$$

$$Mg = mg \sin \theta \implies M = 5(0.5) = 2.5 \text{ kg}$$

Sol 10. Correct Option: (c)

 The movable pulley provides a mechanical advantage where two segments of string pull upward with tension F . Net force on mass M :

$$\sum F = 2F - Mg = Ma \implies a = \frac{2F}{M} - g$$

Sol 11. Correct Option: (d)

 Let F be the pull force. Two rope segments pull the (man + cradle) system up:

$$2F - (M + m)g = (M + m)a$$

$$2F - 80(10) = 80(2) \implies 2F = 800 + 160 = 960$$

$$F = 480 \text{ N}$$

Sol 12. Correct Option: (a)

 Drawing FBD of the man: upward forces are normal force N and hands' pull reaction F ; downward force is Mg :

$$N + F - Mg = Ma$$

$$N + 480 - 600 = 120 \implies N = 240 \text{ N}$$

Sol 13. Correct Option: (b)

Net accelerating force along string:

$$F_{\text{net}} = m_2g \sin 60^\circ - m_1g \sin 30^\circ$$

$$F_{\text{net}} = 10(10) \frac{\sqrt{3}}{2} - 5(10) \frac{1}{2} = 50\sqrt{3} - 25 \text{ N}$$

$$a = \frac{50\sqrt{3} - 25}{15}$$

$$a = \frac{5(2\sqrt{3} - 1)}{3} \approx 4.11 \text{ m/s}^2$$

Sol 14. Correct Option: (c)

Resolving horizontally: $T_1 \cos 30^\circ = T_2 \cos 60^\circ \Rightarrow T_1 \frac{\sqrt{3}}{2} = T_2 \frac{1}{2} \Rightarrow T_2 = \sqrt{3}T_1$.
 Resolving vertically: $T_1 \sin 30^\circ + T_2 \sin 60^\circ = mg = 20 \text{ N}$.

$$\frac{1}{2}T_1 + \frac{3}{2}T_1 = 20 \Rightarrow T_1 = 10 \text{ N}$$

$$T_2 = 10\sqrt{3} \text{ N}$$

Sol 15. Correct Option: (d)

In the reference frame of the wedge (pseudo-force ma rightward):
 Balancing along the incline:

$$mg \sin \theta = ma \cos \theta \Rightarrow a = g \tan \theta$$

Directed horizontally to the left (into the incline).

Sol 16. Correct Option: (a)

Normal force from wedge in the non-inertial frame:

$$N = mg \cos \theta + ma \sin \theta$$

Substituting $a = g \tan \theta$:

$$N = mg \left(\cos \theta + \frac{\sin^2 \theta}{\cos \theta} \right) = \frac{mg}{\cos \theta}$$

Sol 17. Correct Option: (b)

At the very instant A begins accelerating, the spring deformation is zero ($\Delta x = 0$). Hence, spring force on block B is $F_s = k\Delta x = 0$, so $a_B = 0$.

Sol 18. Correct Option: (c)

Total force on table consists of: 1. Weight of accumulated chain: $W = \left(\frac{M}{L}x\right)g$. 2. Rate of momentum transfer: $F_v = v \left(\frac{dm}{dt}\right) = v \left(\frac{M}{L}v\right) = \frac{M}{L}v^2 = \frac{M}{L}(2gx) = \frac{2Mgx}{L}$.
 Total normal force:

$$N = W + F_v = \frac{Mgx}{L} + \frac{2Mgx}{L} = \frac{3Mgx}{L}$$

Sol 19. Correct Option: (d)

Length of string $l = x_A + 2x_B + \text{constant}$.
 Differentiating with respect to time:

$$\dot{x}_A + 2\dot{x}_B = 0 \Rightarrow v_B = \frac{v_A}{2}$$

Sol 20. Correct Option: (a)

Let N_w be normal force from vertical wall, N_i from incline.

Balancing vertically: $N_i \sin(60^\circ) = Mg \Rightarrow N_i = \frac{Mg}{\sqrt{3}/2} = \frac{2Mg}{\sqrt{3}}$.

Balancing horizontally:

$$N_w = N_i \cos(60^\circ) = \left(\frac{2Mg}{\sqrt{3}}\right) \left(\frac{1}{2}\right) = \frac{Mg}{\sqrt{3}}$$

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